

 CASE STUDY PRESENTATION

Thermal Stress & Freezing Horizons

Climatological Predictive Modeling via Heathrow ECA&D Observations



Classifiers



17,155 Daily
Records

PROJECT OVERVIEW

TARGET AREA A: HEAT STRESS

Heatwave Forecasting

Forecasting heatwave occurrence by predicting daily maximum temperatures and identifying periods of sustained exceedance.

Business Impact: Provides advance warning of heatwave events, supporting preparedness, resource allocation, and heat-risk mitigation.

TARGET AREA B: COLD STRESS

Freeze-Risk Classification

Modeling complex daily interactions to determine when average daily temperature drops below 3°C.

Business Impact: Logistics routing to cope with severe frost threats.

DATA SELECTION & PREPROCESSING

ECA&D Weather Parameters:

Our source dataset aggregates 10 key continuous parameters measured at Heathrow (Station ID: 1862) from 1979 to 2025.

 **Temperatures:** Maximum (TX), Minimum (TN), Mean (TG)

 **Atmospheric:** Relative Humidity (HU), Cloud Cover (CC), Rain (RR)

 **Dynamics:** Sea Level Pressure (PP), Sunshine (SS), Snow Depth (SD)

Cleaning Pipeline Step	Resolution Logic Applied
Invalid Flag Interpolation	Values where QA flag is 9 or -99.9 are replaced and linearly interpolated.
Leap Day Removal	Dropped February 29th across years to keep clean, uniform 365-day sequences.
Anomalous Filtering	Identified and removed “suspect” data after determining context (e.g. mean daily temp > max daily temp)

SEASONAL BASELINE & LAG FEATURES

TX Baseline

Heatwave Threshold Calculation: Computing the 90th-percentile threshold from the 1991–2020 reference period using a ± 5 -day moving window centered on each calendar day .

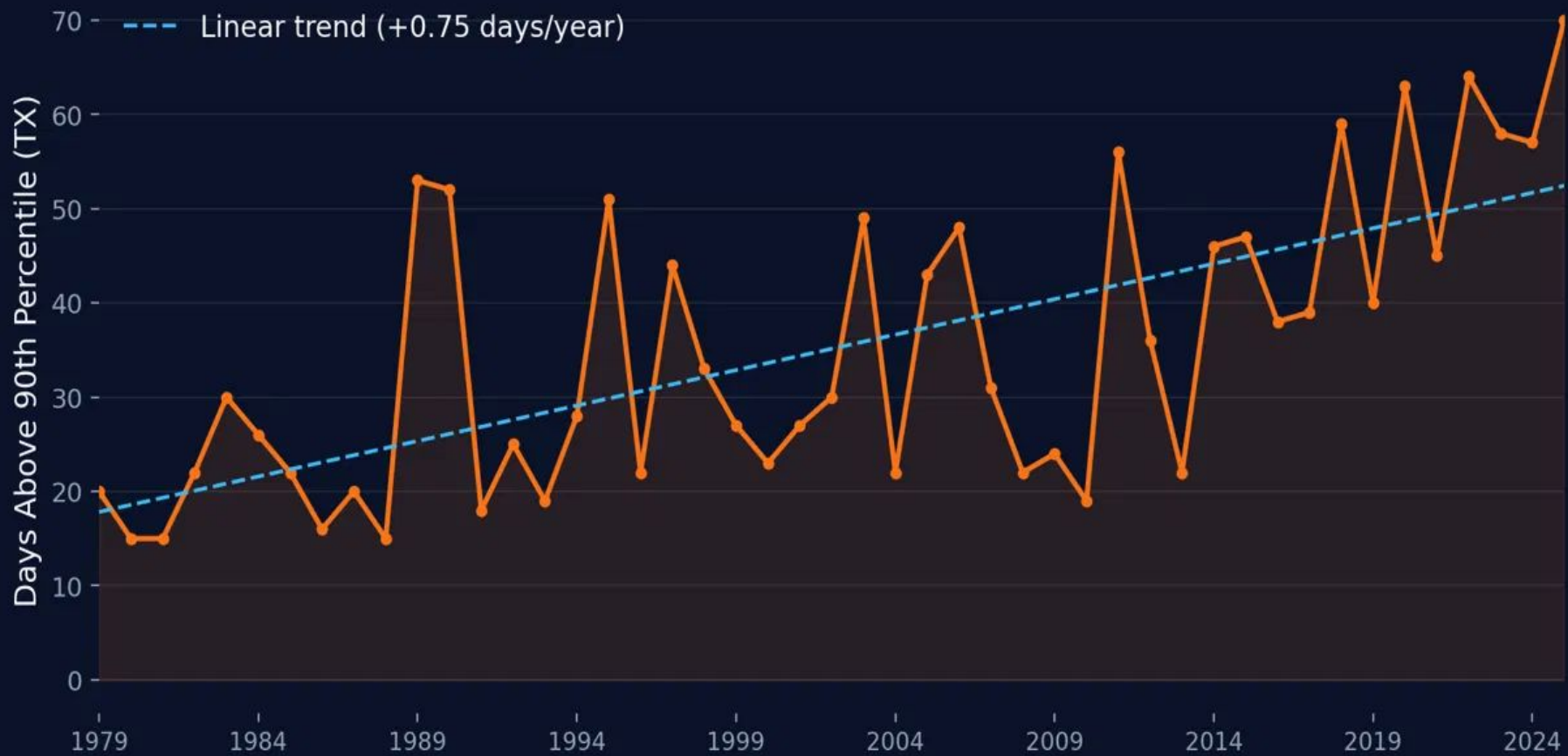
The Heatwave Definition: A heatwave is defined as a period of **three or more consecutive days** with observed temperatures exceeding the calculated daily 90th-percentile threshold in the seasonal May–September window .

Lag Features at Forecast Time

Forecast Initialization Data: Lagged temperature variables were calculated from the historical dataset and included as additional predictor features.

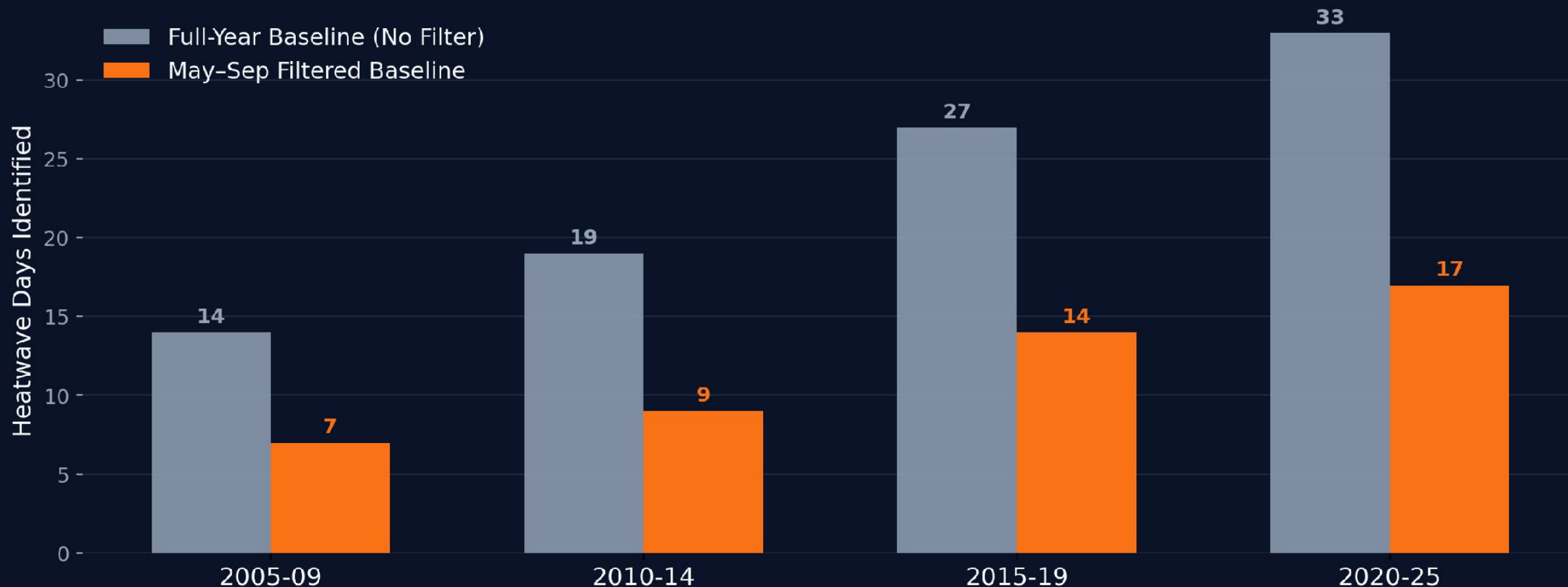
Specifically, temperatures from previous days (e.g., 1-, 3-, and 7-day lags) were used to capture temporal persistence in temperature patterns.

BASELINE 90TH PERCENTILE HEATWAVES



Annual count of days exceeding the 90th-percentile maximum temperature threshold, full calendar-year baseline (1991–2020 reference), Heathrow 1979–2025

SEASONAL FILTERING DOUBLED FALSE POSITIVES



Unfiltered baseline nearly doubled heatwave counts by absorbing abnormal spring/autumn TX spikes into the percentile calculation

FEATURE SELECTION VIA CORRELATION

Pearson Correlation Matrix: 19 Engineered Features vs Heatwave Target

Feature	Correlation (r)	Signal Type	Decision	Window
tx (Daily Maximum Temp)	0.26	Strong	Kept	±0d
tx_lag1 (Previous Day Max)	0.23	Strong	Kept	±1d
ss (Sunshine Duration)	0.18	Moderate	Kept	±0d
sd (Snow Depth)	-0.02	Negligible	Dropped	—

Daytime maximum temperature dominates the signal; atmospheric variables (pressure, humidity, cloud cover) showed near-zero linear correlation and were dropped entirely.

FOUR-MODEL HEATWAVE CLASSIFIER COMPARISON

Model Comparison: Heatwave Class Metrics on Held-Out Test Set

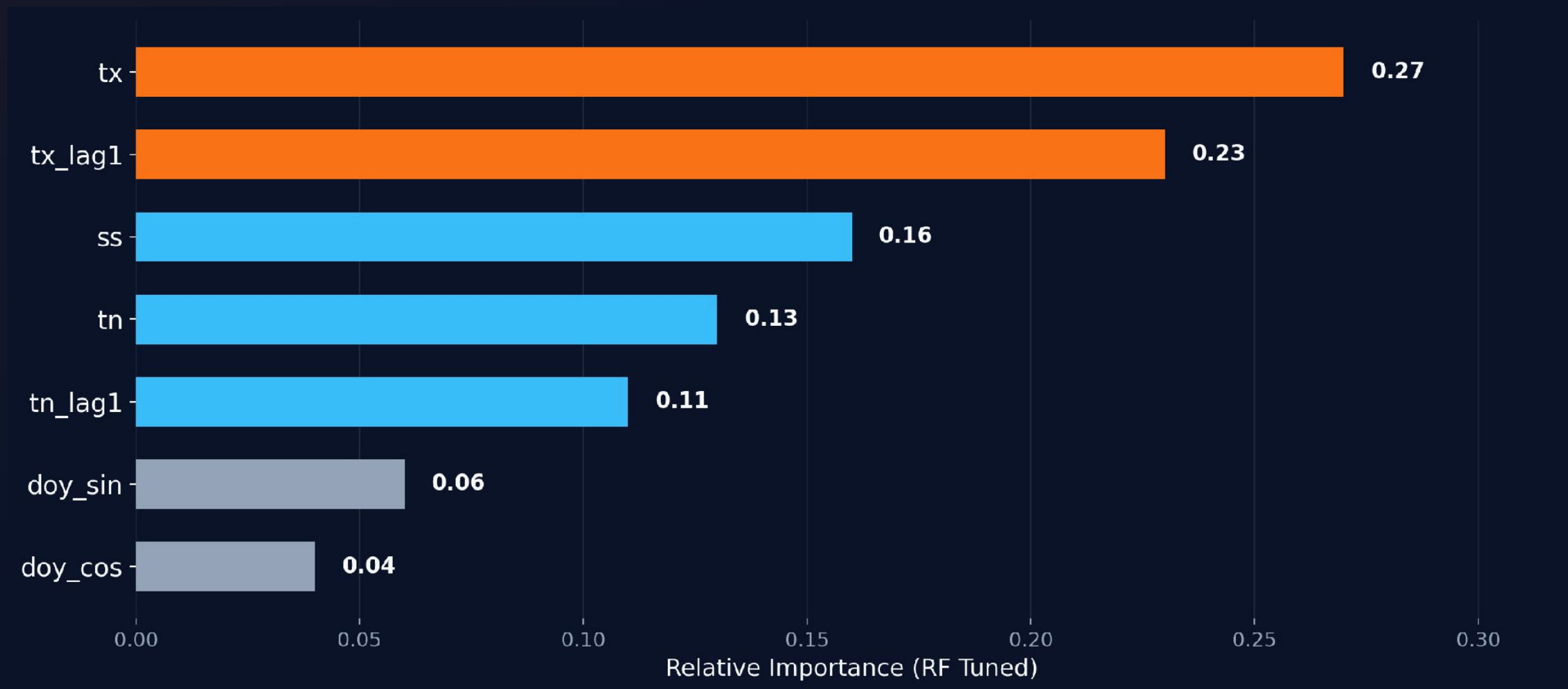
Model	Precision	Recall (Sensitivity)	F1-Score	Notes
KNN (Baseline)	0.775	0.454	0.572	Distance-based, scaled
Random Forest (Class-Weighted)	0.835	0.523	0.643	class_weight=balanced
RF + Oversampling (Best Overall)	0.841	0.667	0.744	Resample minority class
RF Tuned (GridSearchCV)	0.680	0.793	0.732	Highest recall

RF + Oversampling achieved the best balanced F1 (0.744); RF Tuned traded precision for the highest recall (0.793), catching more real heatwave events.

MODEL COMPARISON



WHAT THE MODEL ACTUALLY LEARNED



Feature importance from the tuned Random Forest classifier: same-day maximum temperature (tx) and the prior day's maximum (tx_lag1) jointly account for half of total predictive weight.

LIVE 7-DAY HEATWAVE OUTLOOK

Live Forecast Application: RF + Oversampling, 7-Day Outlook (18–24 June 2026)

Forecast Date	TX (°C)	TN (°C)	Heatwave Prob.	Prediction
Thu 18 June	24.0	16.9	0.00	No Heatwave
Sun 21 June	29.7	19.4	0.25	No Heatwave
Tue 23 June	33.9	24.4	0.35	No Heatwave
Wed 24 June	32.2	24.5	0.37	No Heatwave

No single day crosses the 0.50 decision threshold, but heatwave probability triples across the week as overnight temperatures climb — an early-warning signal worth flagging operationally.

VALIDATION STRATEGY & METRICS

Chronological Data Splits

- **Training Fold:** All historical daily steps up to and including 2021 (15,685 records).
- **Test Fold:** Unseen modern timeline sequence from 2022 through 2025 (1,460 records).

Evaluation Metric	Mathematical Target Focus
Recall / Sensitivity	Maximizes active freeze event identification, minimizing missed hazard risks.
F1-Harmonic Score	Provides a balanced benchmark metric to evaluate model accuracy on highly skewed datasets.

FEATURE ENGINEERING & SELECTION



Cyclical Temporal Vectors

Transformed raw "Day of Year" (DOY) integer counts into cyclical projections:

```
sin(2π * DOY / 365.25)  
cos(2π * DOY / 365.25)
```

Ensures Dec 31st and Jan 1st are handled as adjacent steps rather than polar opposites.



Chronological Lag Matrices

Constructed temporal offset histories for key features to capture memory windows:

- **TN Lag Vectors:** tn_lag1, tn_lag3, tn_lag7
- **TG Lag Vectors:** tg_lag1, tx_lag1, pp_lag1



Predictive Target Creation

Created forward-looking target variables for predictive models:

```
y10: t_mean.shift(-10) < 3  
y5: t_mean.shift(-5) < 3
```

FEATURE SELECTION VIA CORRELATION

Pearson Correlation Matrix: 28 Engineered Features vs Cold Target

Feature	Correlation (r)	Signal Type	Decision	Window
tx_lag1 (Previous Day Max)	-0.45	Moderate	Kept	±1d
tn_lag1 (Daily Minimum Temp)	-0.43	Moderate	Kept	±1d
doy_cos (Day of year, adjusted)	0.32	Moderate	Kept	±0d
ss (Sunshine)	-0.12	Low	Dropped	±0d

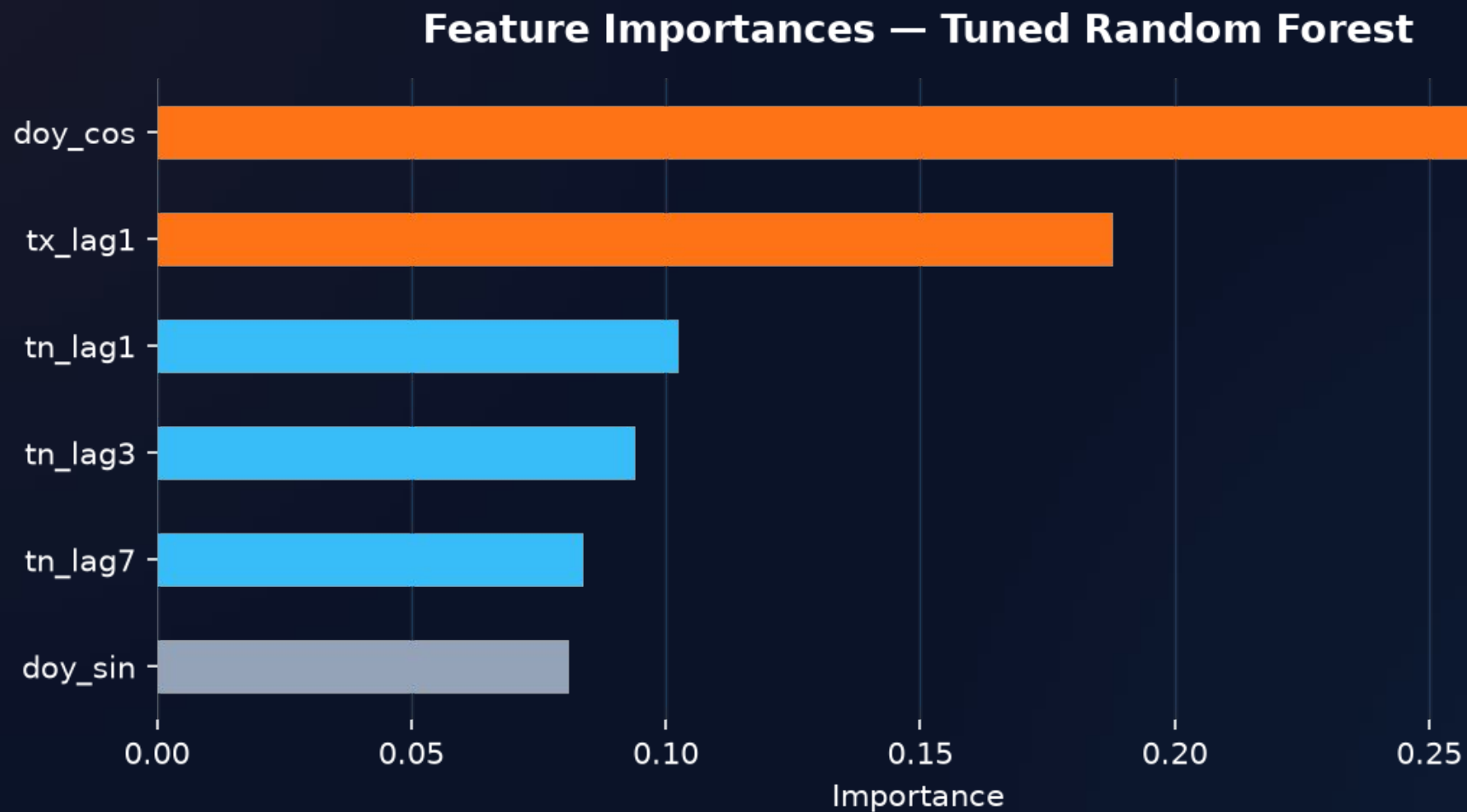
Temperature values have the highest correlations; day of the year had a strong correlation showing that temperatures may be cyclic; conservative dropping strategy.

10 DAY PREDICTION ACCURACY

Model Comparison: Cold Snap Metrics on Held-Out Test Set

Model		Precision	Recall (Sensitivity)	F1-Score	Notes
Random Forest (Class-Weighted=balanced)	True	0.98	0.87	0.93	Misses more cold days, too conservative
	False	0.21	0.70	0.33	
RF + Oversampling	True	0.96	0.99	0.97	Predicts everything as cold, high FP
	False	0.14	0.05	0.07	
RF Tuned (RandomSearchCV)	True	0.97	0.97	0.97	Best Overall, most balanced
	False	0.26	0.22	0.24	

10 DAY PREDICTION – WHAT MATTERED MOST?



Feature importance from the tuned Random Forest classifier: day of the year and the prior day's maximum (tx_lag1) jointly account for nearly half of total predictive weight.

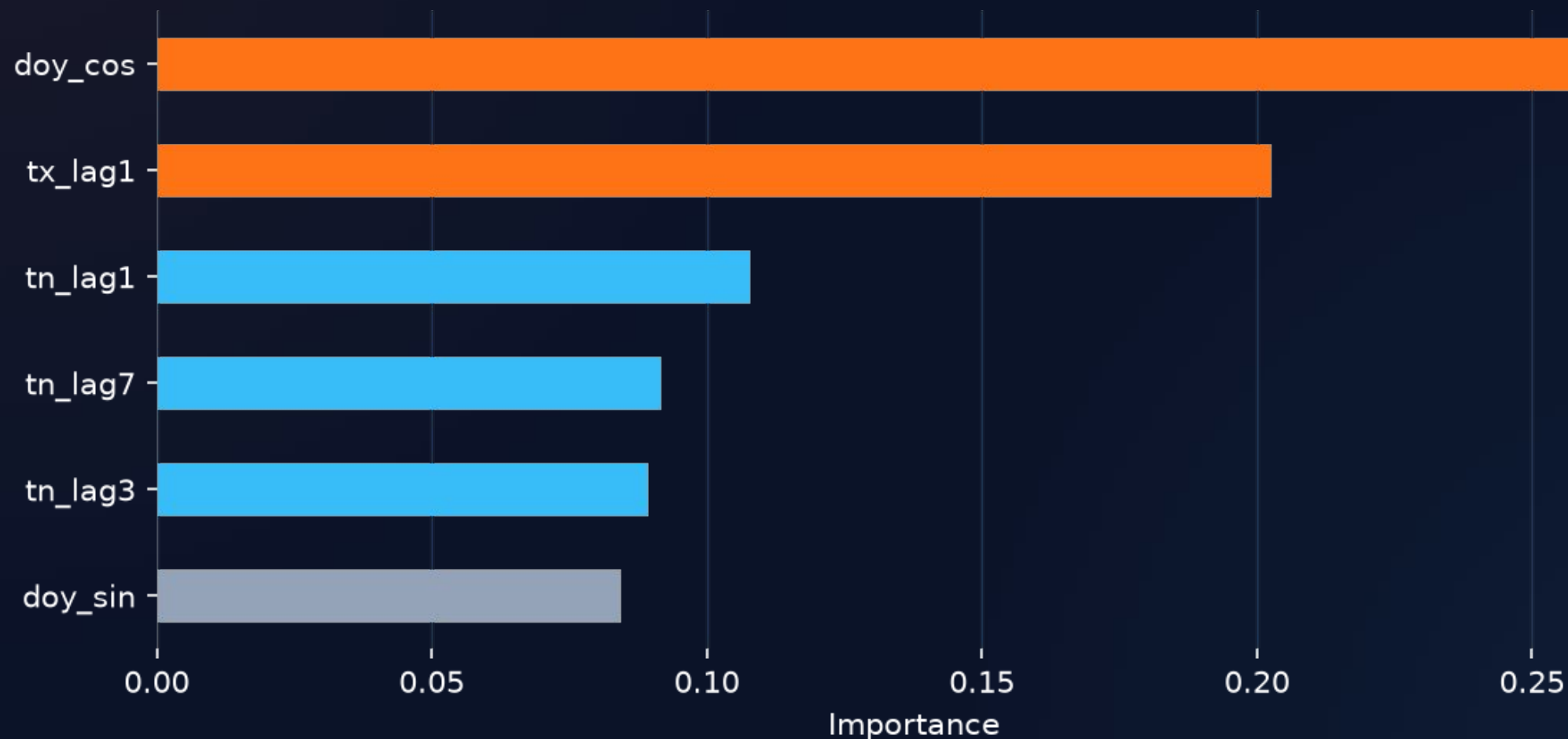
5 DAY PREDICTION ACCURACY

Model Comparison: Cold Snap Metrics on Held-Out Test Set

Model		Precision	Recall (Sensitivity)	F1-Score	Notes
Random Forest (Class-Weighted= balanced)	True	0.98	0.92	0.95	Good balance
	False	0.25	0.63	0.36	
RF + Oversampling	True	1.0	0.80	0.89	Worst recall, misses too many cold days
	False	0.18	0.94	0.29	
RF Tuned (RandomSearchCV)	True	0.96	0.99	0.97	Best Overall, near perfect prediction cold
	False	0.17	0.06	0.09	

5 DAY PREDICTION – WHAT MATTERED MOST?

Feature Importances — Tuned Random Forest



Feature importance from the tuned Random Forest classifier: day of the year and the prior day's maximum (tx_lag1) jointly account for nearly half of total predictive weight.

REAL-WORLD OPERATIONAL IMPACT

- ✓ **Infrastructure Planning:** Early warnings enable transportation and municipal services to prepare for heatwaves and freezing conditions before disruptions occur.
- ✓ **Supply Chain Management:** Advance forecasts support inventory planning and resource allocation, reducing operational disruptions during extreme weather events.
- ✓ **Energy Grid Preparedness:** Predictive weather alerts help utilities anticipate demand fluctuations and prepare capacity for temperature extremes.

Thank you!

Predictive Modeling of Climatological Extremes

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